# Rldcoin: Permissionless Mining and Causal Interstellar Payments

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**Abstract.** Rldcoin proposes a peer-to-peer payment system whose participants may be separated by interstellar distances, prolonged disconnection and one-way contacts. Ordinary users must be able to validate, construct blocks and mine without an account, stake, permit, designated pool, founder service or planetary authority. The design combines one conserved proof-of-work currency with portable owner-signed payment intents, independently replaceable verification evidence and authenticated store-and-forward carriage. A payment can travel while its sender is offline; a recipient executes the evidence locally and can authorize a subsequent payment. Reconnection preserves and reconsiders still-valid signed intents, while conflicting spends and rewards from displaced histories remain reversible. This is causal, conditional payment: communication cannot outrun light, local knowledge cannot establish the absence of stronger unseen work, and copyable software cannot guarantee conflicting isolated payments are both irrevocable. This paper defines required capabilities, explicit limits and falsifiable release gates. It establishes neither a monetary deployment nor an operational interstellar route.

## 1. Purpose and scope

The primary requirement is replaceability: no founder machine, server, validator identity, directory or geographic origin is required for ordinary mining and payments. If only one eligible user node remains anywhere, it must not be refused solely because other participants are absent. It can construct valid transactions and attempt proof of work using its retained, independently validated history. Its rewards and payments remain conditional on that history being selected when other histories become available.

The product goal has two independently necessary parts. Open participation gives an ordinary person a direct path to verification and solo mining on supported general-purpose hardware. Interstellar payment gives an owner a direct path to transferring authorized value and its validation dependencies across real, delayed contacts without keeping the sender, founder or Earth online. A chain that only admits privileged miners fails the first goal; a packet relay that never reaches independently verified recipient spendability fails the second. Both must be tested through ordinary commands and owner-controlled wallets.

The architectural contribution is the separation of durable owner intent, transport custody and locally executed payment state. Consensus selects money; contacts move exact bytes; a recipient chooses whether the resulting conditional position is acceptable. These roles may be performed by different devices and at different times. Neither a fast local network nor a slow stellar route changes who may authorize a payment, create a subsidy or decide that a conflicting input was spent. This is a design composition, not a claim of priority over existing proof-of-work, delay-tolerant networking or causal-transfer research.

A node cannot distinguish the last surviving participant from one isolated by a partition or an attacker. Accordingly, the protocol does not promise a fixed block time, guaranteed mining income, absolute payment finality, or preservation of all disconnected branches. An offline recipient can be named in a transaction, but has not independently verified receipt. A wallet signature cannot be produced without its owner's authority, even if only one node remains.

The revision replaces the previous design's mandatory regional validator quorum, irreversible regional export/import baseline and three/five-second final-spendability target. It retains the monetary cap, zero initial allocation, deterministic issuance recurrence, owner authorization, exact accounting, explicit cryptographic versions and evidence-based release requirements. Existing signed networks and historical results are not upgraded by publication.

The companion Survivor-1 contract fixes consensus bytes and monetary rules [6]. The additional interstellar open-payment requirements define the product, evidence-carriage and reconsideration obligations in this paper [7]. They do not silently change Survivor-1 consensus or supply a new capsule wire codec. Any new codec or work algorithm needs its own exact version, bounds, public vectors and adoption review before use. Required behavior belongs here; implementation progress belongs in the separate implementation plan.

## 2. One currency and one selected history

A fresh genesis binds the currency identity, zero issued supply, protocol and suite versions, network kind, network discriminator, start time and initial target. The exact profile is RLD-SURVIVOR-1, specified in the companion consensus, operations and conformance documents [6]. Its published identity is a common starting point, not a permanently online issuer or permission service. Independent implementations use the same consensus domains; an implementation source hash is release evidence rather than a consensus identifier. A node verifies from genesis and does not accept balances supplied by an operator, relay or serialized cache as authority.

The baseline uses one proof-of-work history per currency. Earth, a spacecraft and another stellar system describe locations of participants, not separate issuance authorities. Any eligible miner can extend the same currency's history from any location. A partition can produce competing branches, but their rewards and balances are never added together. Replaying the selected valid branch determines issuance and spendable state.

This choice avoids inventing new money whenever a region becomes disconnected. It also means that independently growing branches do not provide permanent regional settlement autonomy: a branch may later lose selection. Independently final regional ledgers, bridges and cross-region minting rights are outside this baseline and require a separate conservation and reorganization design before adoption.

## 3. Mining, validation and chain selection

Mining is permissionless under the adopted rules. Ordinary block validity must not require approval from the founder or a fixed signer committee. A non-mining full node still verifies and relays; running a node alone earns no subsidy. A miner earns only the subsidy and permitted fees of valid blocks retained in the selected history, subject to maturity and later reorganization.

Permissionless has operational meaning. A conforming user must need no registration, identity check, invitation, payment to an operator, prior RLD balance, stake, validator seat, manufacturer attestation, subscription, remote API credential or mandatory pool. The ordinary client must construct and validate its own block template and let the user choose a valid reward key. Mining can be enabled, bounded and stopped locally without surrendering a spending key. A pool or hosted service is optional; its failure cannot disable independent validation or solo work. Public rules and resource prerequisites apply equally to founder and newcomer. A neighbor's willingness to carry bytes is separate from mining eligibility; no protocol can compel an unwilling radio operator to provide a contact.

General-purpose hardware is a concrete release obligation, not an assertion that every powered device is adequate. The supported envelope must include ordinary ARM64 and x86-64 computers mining without a specialized accelerator. Publish reproducible installation, full-validation and solo-mining measurements with RAM, disk, energy, thermal behavior, verification cost and interactive usability. Users control worker and power budgets locally; those controls cannot alter the network's required work. The same exact header and work must be checked on both architectures. A release must explain what a measured work rate implies at the actual selected difficulty, including potentially negligible reward prospects.

Open entry does not give one-person-one-vote or equal income. Software cannot distinguish one person with many machines from many people. Double-SHA256 permits general-purpose computation but also specialized-hardware advantage. The baseline does not promise ASIC resistance, profitable laptop mining or equal rewards. A future CPU-oriented or memory-hard work profile must fix exact code-independent semantics, parameters, key derivation, cross-architecture vectors and hostile verification costs; its name alone is no security argument. It cannot silently reuse the current work score or upgrade existing value.

Every accepted block must satisfy its parent linkage, difficulty, proof of work, canonical encoding, size and resource bounds, owner signatures, input uniqueness, maturity and exact issuance rules. An invalid branch never wins, regardless of claimed work. For target T, each block contributes floor(2^256 / (T+1)); genesis contributes zero. A full node recomputes cumulative work from validated targets rather than trusting a peer's score. Among complete eligible valid branches it selects the greatest cumulative work; equal work selects the lexicographically smallest tip block ID. Peer count, geography and claimed uptime never confer weight. No confirmation-depth checkpoint prevents a valid deeper reorganization.

A reorganization removes the old suffix and executes the new suffix atomically. Displaced rewards cease to exist in the selected state. Displaced payments may be reconsidered only if their inputs and signatures remain valid; conflicting payments do not survive by combining balances. Wallets must report changed confirmations and conflicts. A backup, relay acknowledgment or previous local acceptance cannot override the selected ledger.

Security depends on actual work, communication and validation. A lone miner can continue computation but offers no independent resistance to a stronger competing history. A small or isolated network must disclose its exposure to withholding, eclipse, double-spend and majority-work attacks. Confirmations observed under isolation are not an estimate of unseen honest work.

## 4. Difficulty and single-node operation

A proof-of-work network permits a single miner to try; it does not guarantee that the miner can solve the current target promptly. For stable work rate H and required expected work W, expected block time is W/H, not a deadline. If H falls sharply, the next block can be much later than the nominal interval. There is no finite worst-case discovery time.

Survivor-1 fixes double-SHA256 header work, initial and maximum target 2^248-1, nominal spacing 600 seconds, and a 144-block adjustment window. At a parent height divisible by 144, the base target scales by the timestamp span of the preceding 144 mined blocks divided by 85800 seconds; the span is clamped to 21450 through 343200 seconds and the resulting target to 1 through 2^248-1. Other blocks inherit the parent's base target. A child timestamp must exceed its parent's timestamp and the median of up to eleven ancestors. Local activation defers a block more than 7200 seconds ahead of a reliable local clock; delayed old evidence has no maximum delivery age. All arithmetic and byte encodings are fixed by [6].

When the child timestamp is at least 14400 seconds after its parent, the same public consensus rule permits the maximum target. The recovery block receives only the work of that actual target; it does not inherit the harder base score or reset base difficulty. No peer-count claim, founder approval, quorum or emergency key is involved. At that maximum target a uniformly distributed header succeeds with probability 1/256 per distinct trial. This preserves a probabilistic path for a weak surviving miner, not a four-hour deadline. Repeated recovery blocks and reward maturity may still take many days. Timestamp manipulation, waiting attacks, work oscillation and withholding require independent analysis; the rule does not inherit Bitcoin's confirmation probabilities.

Single-node acceptance must therefore report eligibility, actual block discoveries, elapsed time, resource use and uncertainty. It must separately test loss of most mining capacity at a previously harder target. Passing an easy fresh-genesis example does not establish useful liveness after a production-scale collapse in work rate.

Recovery is an availability tradeoff rather than a hidden fast-payment claim. If a hard base target leaves a weak miner relying on repeated recovery gaps, 100 such gaps cover 400 timestamp-hours before reward maturity; random discovery adds further waiting. Timestamp gaps are not proof of elapsed physical time, and local future-clock allowance affects eligibility. Tests must include 99% and 99.99% work loss from a retained hard history, future timestamps, withheld recovery branches and early low-difficulty reward concentration. Zero genesis allocation does not by itself establish a fair launch. Public launch notice, exact artifact availability and the distribution of early selected rewards require separate review; there is no privileged founder mining mode.

## 5. Ownership, transfer and confirmation

A payment spends explicitly identified outputs using their owners' signatures. All amounts are checked integers; inputs equal new outputs plus explicit fees. Unknown inputs, duplicate spends, invalid signatures, overflow and immature rewards refuse. Change belongs to the authorized owner. Fees move existing value and cannot expand supply.

Wallets distinguish created, submitted, relayed, included, confirmations observed, locally spendable, conflicted and reorganized. Confirmation count belongs to an exact selected block and observed history; it is neither a delivery receipt nor irreversible finality. A recipient independently validates the payment and its inputs before choosing whether to accept its risk. An actual subsequent payment demonstrates local execution, not immunity to a later reorganization.

The candidate baseline makes a reward created at height h eligible in a spending block at height h+100, including its associated miner fees. Ordinary outputs may be spent only in a later block; same-block child transactions refuse. Reorganization can remove both reward and descendants. One strict Ed25519 owner authorizes each transfer's ordered inputs, outputs, explicit fee and height expiry. The default expiry is the maximum u128 height, avoiding an unintended short deadline for interstellar carriage. User interfaces must not turn local maturity, an acknowledgment, cached balance or connectivity indicator into a final-settlement claim.

Offline recipients can receive an output addressed to their key without signing the receipt. They cannot independently confirm it until evidence reaches their node. Losing all valid history, necessary software, keys or resources is outside the single-surviving-node premise. No surviving node may spend somebody else's output without authorization.

An intent must survive a change in its carriage. The owner signs exact currency, input outpoints, ordered output owners and amounts, fee and height expiry. A relay may supply another valid inclusion history, but may not change any signed field, substitute a key or increase the fee. A payment's first submission time and signed bytes remain retained through retries. A destination address is sufficient for making the output; a return signature, account registration or reachable receiver service is not a prerequisite to signing it.

After reorganization, durable wallet and pool policy reconsider the original transaction in dependency order. If its exact inputs still exist, belong to its signer, are mature and unspent, and its original expiry permits inclusion, the same signature may be offered again. Dependencies between ordinary outputs require successive blocks under the parent-only rule. Reconsideration must not double-reserve funds or silently first-sign a replacement. A missing dependency waits; an input spent elsewhere is conflicted; an expired intent is expired; a reward removed with its branch and every dependent payment remain unsupported by that state. Retain the original intent and its cause, even when it cannot be reoffered. Inclusion still depends on valid inputs, capacity, fees and miners' choices.

For example, Alice's input already exists in the common history before a long partition. Her unchanged signed payment to Bob can be reoffered after reconnection if that exact input survives unspent. If Alice instead spends a reward created only on the losing branch, neither Bob's receipt nor his subsequent signature can recreate that reward. Non-conflicting owner intentions can be retained; non-conflicting intentions are not necessarily backed by surviving money. This distinction prevents evidence preservation from becoming compensation issuance.

## 6. Supply and issuance

The maximum supply is 100,000,000,000 RLD, exactly C = 10^35 runlai, with 1 RLD = 10^24 runlai. Genesis height 0 issues nothing and allocates nothing. The reserve belongs to the currency's deterministic schedule, not to the founder or to a planet. Any eligible miner may compete for the next selected block's reward; regional labels grant no extra reserve.

For height h >= 1, let L = 200000, era j = floor((h-1)/L), and slot k = (h-1) mod L. At era start R_j is the remaining reserve. If R_j >= 2, B_j = floor(R_j/2); otherwise B_j = R_j. Let q_j = floor(B_j/L) and m_j = B_j mod L. The reward is q_j + 1 runlai when k < m_j, otherwise q_j. A completed era leaves R_(j+1) = R_j - B_j. All arithmetic is checked integers. An incomplete era issues only rewards from its selected blocks. Reorganization recomputes the reserve from the replacement history; competing branches never accumulate independent supply.

```
EraBudget(R):
  require 0 <= R <= 10^35
  return R if R < 2 else floor(R / 2)
Reward(R, k):
  require 0 <= k < 200000
  B = EraBudget(R)
  q, m = divmod(B, 200000)
  return q + (1 if k < m else 0)
```

Boundary vectors retain the original recurrence. With R_0 = 10^35, heights 1 through 200000 each issue 250000 RLD; height 200001 issues 125000 RLD. For R = 3, 2 or 1, slot 0 issues one runlai and the remaining slots issue zero; the next reserves are 2, 1 and 0. For R = 0 all rewards are zero. For R = 400003, slot 0 issues two runlai and every other slot issues one. The final one-runlai reserve is released once. No calendar, service receipt or compensation claim creates additional issuance.

For each fully replayed selected history, cumulative issuance equals the sum of all unspent outputs, including immature rewards, and never exceeds C. Fees are counted once through their recipient outputs. No baseline bridge reserve, imported duplicate balance or off-ledger compensation is allowed. Any future escrow extension must define mutually exclusive accounting and reorganization behavior. A supply cap does not guarantee price, liquidity, fair distribution or a funded long-term security budget.

## 7. Interstellar communication and payment

The transport target is authenticated, bounded store-and-forward. Nodes learn reachable neighbors, retain admitted transactions and evidence durably, and forward when a contact exists. Normal startup should support relay without a mandatory Earth directory. Bootstrap requires some prior trusted currency information and an actual contact or carried archive; discovery cannot manufacture connectivity.

A portable payment capsule is a logical collection of an exact signed transaction, its currency identity, a bounded manifest of validation dependencies and observations of particular inclusion histories. It is not a new coin, a destination mint or an authority certificate. The owner signature remains unchanged when later evidence is attached. The transport authenticates origin, destination and exact carried bytes; the node executes the transaction under consensus. Before an implementation offers a capsule codec, that codec must define canonical bytes, versions, manifest ordering, complete-byte commitments, chunk reassembly, size and quota bounds, and cross-implementation negative vectors. Calling a message a capsule grants no new accepted wire format.

Evidence resolves to complete native-validated history from the pinned genesis or an exact predecessor already fully executed by the recipient under the identical rules. A manifest, state root, work claim, header chain or compressed representation cannot initialize a trusted balance. Deduplication may share only exact authenticated bytes while retaining their necessary dependencies. Missing material is requested or carried later and remains unresolved. Admission bounds and interrupted assembly must preserve earlier admitted obligations. A sender cannot hide the full-history cost by reporting only a small transaction or manifest.

Carriage is contact-first rather than request-response-first. Ordinary nodes must proactively offer missing history and payment bytes over a configured one-way path even if no reverse request or receipt can return. Evidence is offered dependency-first, in bounded fair batches, without letting one indefinitely missing ancestor or one large payment starve all unrelated admitted work. The sender may leave after durable custody is accepted by a willing relay; this transfers a byte-retention obligation, not a guarantee of destination delivery or monetary settlement. A destination receipt may travel later by another path. Permanent receipt loss remains unknown to the sender and must not block otherwise valid recipient execution.

A relay identity, route advertisement, packet digest or custody receipt grants no ownership or issuance authority. Survivor-1 defines signed origin packets, pinned neighbor identities, interactive challenge-bound delivery, offline one-way frames and signed destination-receipt return packets. Ground interactive links use pinned TLS 1.3 without downgrade. Origin identity, exact destination and bytes remain authenticated across forwarding; there is no claim of a complete signed physical path. Duplicate packets require exact-byte checks and durable custody before acknowledgment.

Default carriage admission is 4096 packets and 256 MiB, separate from ledger history growth. Capacity refusal preserves existing obligations. A hop receipt alone cannot delete the last addressed copy; a valid destination receipt or explicit withdrawal is required. Message expiry never refunds a payment.

Neighbor discovery may describe signed candidate routes, but an advertisement must not choose a node's trusted genesis, signing authority, connection endpoint or certificate pin. Admission to a privately operated link is explicit and locally controlled; a missing pin or unreachable first contact is a communication limitation, not loss of the right to mine. Mobile carriers, removable archives and future radio or optical adapters must preserve exact-byte authentication, quotas and custody across changes of medium. BPv7 or BPSec interoperability requires a specified adapter and its own vectors; mentioning those standards is not implementation or route qualification [3,8].

Cross-stellar payment in the baseline means carrying a transaction and enough authenticated history for the recipient to validate it on the same currency. It is not a separate destination mint. The receiving node must reconcile the received branch with its own valid history before reporting confirmations or spendability. Missing ancestors remain unresolved. An isolated sender's claim of greatest work is not proof that no stronger history exists elsewhere.

The earliest independent receipt is bounded by actual propagation and contact availability; prolonged or permanent disconnection can make delivery indefinitely pending. NASA's delay-tolerant networking and Bundle Protocol provide relevant transport techniques [2,3], but do not prove monetary finality, key longevity or route availability. Physical contacts, operators and storage must be qualified separately for each advertised route.

Each advertised route needs a predeclared useful operating envelope: directed contacts and outages, payload capacity, storage, replay CPU, supported history length and divergence, and the maximum acceptable pending exposure chosen by the user. A fresh destination may need the whole relevant chain; evidence growth can outrun contact capacity. Measure total dependency bytes, retries, backlog and actual onward payment, including failures. A route with delivered packets but no independently spendable recipient output has not demonstrated payment usefulness. Earth-to-Mars, Earth-to-nearest-star and a spacecraft-local network require separate envelopes; topology drawings are neither physical links nor evidence of viable monetary settlement.

The payment interface must bind each observation to an exact transaction and selected tip, observation time, inclusion position, validated work after inclusion, known competing validated heads, missing history and locally observed contact history. Unseen competing work is UNKNOWN. Contact age, signed peer status and confirmation count cannot prove global freshness or yield a justified numeric attack probability by themselves. A recipient may set a local amount or exposure limit and decline a conditional position. Such a policy changes that user's acceptance, not issuance, mining eligibility or consensus validity. A receipt received today still describes the exact historical event it authenticates.

The ordinary interstellar flow is deliberately asymmetric:

```
Owner signs exact intent and retains its reservation
  -> willing relays carry intent and complete dependencies
  -> recipient validates history and selects a local branch
  -> recipient sees conditional balance and its evidence gaps
  -> recipient signs a real onward payment when locally eligible
Return receipts may follow later; silence never credits or refunds.
```

Long interstellar partitions can create deep competing histories. When contact resumes, cumulative-work selection may displace years of a weaker branch. The baseline therefore supports delayed transport and conditional payment verification, not guaranteed preservation of autonomous stellar economies. Useful interstellar settlement under realistic work distributions and latency remains an open acceptance problem. Merely increasing confirmation counts cannot remove this limitation.

## 8. Keys, archives and upgrades

Private keys remain with their owners. The candidate defines native encrypted wallet key files using fixed scrypt parameters and AES-256-GCM; a key file alone is not a complete wallet backup. Reviewed intent and input reservations are durable before signature release; exact retries recover the original signed transfer rather than first-signing again. Reorganization restores unresolved reservations when inputs reappear. Restore must validate full history and retained operations; an old backup cannot prove its own freshness. Without independently retained freshness evidence, the wallet discloses FRESHNESS_UNKNOWN and requires explicit local risk acknowledgment before signing; mining with an existing public reward key remains possible. Direct-key loss has no protocol reset or administrator compensation. Cross-device custody and hardware failure still require independent testing.

A usable survivor needs recoverable data. Nodes must specify archive retention, resource ceilings, pruning dependencies, full replay and recovery behavior. A digest identifies bytes; it cannot substitute for execution or recreate missing evidence. Test history beyond the first 200000-block issuance era, long partitions and deep reorganization within explicitly declared storage and CPU bounds. If an implementation cannot safely reconcile a history within those bounds it must refuse the affected operation and disclose the limitation, not fabricate acceptance.

Protocol, work algorithm, signature suite, hash and key-era identifiers must be explicit. Upgrades require public rules, compatible history interpretation and independently verifiable adoption. Unknown or expired cryptographic assumptions cannot authorize new acceptance. Post-quantum signatures, renewal and long-delayed revocation remain separate engineering and review obligations; their presence in a component does not qualify monetary use.

No finite cryptographic or storage test establishes a hundred-million-year guarantee. Long continuity requires funded maintenance, generational succession, media refresh, authenticated renewal and finite periods of requalification. Loss beyond recoverable evidence remains loss, not permission to issue replacement currency.

## 9. Threats and limitations

The principal threats are dominant or withheld work, network eclipse and partition, double spending, timestamp and difficulty manipulation, invalid-history flooding, storage exhaustion, key compromise, rollback, software supply-chain compromise and misleading payment displays. Every release must state which assumptions and attack budgets its evidence covers. Node count is not a measure of independent work or control.

Protecting ordinary participation also requires bounded work before expensive authentication, fair local service, disk admission, and isolation of contact pressure from local mining and owner review. These are availability defenses rather than membership gates. Every complete accepted transaction, block and received envelope still needs its required verification. A process-local cache, reputation score, puzzle or rate limit must not turn unchecked evidence into value or make an operator's identity the consensus admission rule. Authenticated peers can still send costly valid material; the declared hardware envelope must include hostile admitted load.

An attacker may reorganize payments using stronger valid work; it cannot make an invalid signature or excessive subsidy valid. Economic loss can still occur even when all protocol rules execute correctly. Risk grows when a recipient cannot compare histories, when work is concentrated, or when a payment is accepted before competing work can propagate. Privacy is not anonymity: transaction graphs, contact metadata and addresses can reveal relationships.

The design cannot simultaneously provide unrestricted progress in every isolated partition, one consistent spend history and irrevocable acceptance of conflicting payments. This revision chooses continued local attempts and explicit reorganization. It does not silently inherit the older committee-finality or regional export/import guarantees. Optional bridges, channels and independent regional settlement require separately reviewed rules and cannot be advertised as baseline features.

The smallest counterexample uses one software-owned input copied before spending. A malicious owner signs one payment to Bob and another to Carol, whose isolated nodes cannot communicate. If both must accept irrevocably using only their local evidence, both can accept the same predecessor. Hash chains, local counters, signatures or extra work do not prove that the other copy never existed. Later conflict detection, a deterministic winner or insurance cannot make

both earlier receipts irrevocable and conserved. The exclusive-rights research contract makes the assumptions and conditional alternatives explicit [9].

Causal ledgers and disjoint issuance rights remain promising research directions for preserving independent stellar economies. They require a concrete non-equivocation mechanism, open and Sybil-resistant allocation of every issuance slot, authority succession, complete causal evidence and exact conservation. An ideal single writer is an assumption, not a solution for copyable ordinary machines. An intersecting quorum can exclude conflicts under authenticated membership and honest-lock assumptions but may be unavailable to a lone survivor. Qualified non-rollbackable hardware adds device trust and replacement obligations. A DAG alone neither prevents duplicate spending nor assigns the unchanged reserve. No such alternative can replace the baseline by relabeling its assumptions or weakening the cap.

## 10. Acceptance and release

G1 - Open participation and survivor operation. An independent user installs the published software, validates genesis/history, creates a wallet and mines without a founder service or fixed signer approval. Shut down the founder and every other fixture node; retain the survivor's original history and target. Demonstrate block discovery, mature reward spending, receipt and an actual second owner-authorized payment. Report failures, waiting times and the exact work assumptions. Repeat with different surviving participants and with a sharp loss of mining capacity.

G2 - Partition and convergence. Separate miners and owners, submit conflicting spends, extend both branches and reconnect. Verify exact cumulative-work selection, equal-work behavior, displaced rewards, transaction reconsideration and atomic cold replay. No invalid history wins and no branch balances are combined. Include eclipse, withholding, timestamps, oscillating work, invalid proofs and resource exhaustion.

G3 - Money and wallets. Verify the cap, zero allocation, all issuance boundary vectors, 100-block reward maturity, fees, checked arithmetic, signatures, backups, reorganization displays and cross-device recovery. Measure submission, inclusion, observed confirmations and recipient acceptance separately. Retain every valid submission and original clock; no success-only performance statistics.

G4 - Delayed transport. Use ordinary nodes and actual authenticated contacts to carry complete evidence through interruption, duplication, reordering, corruption, limited capacity and restart. Distinguish transport custody from locally accepted money. Test long logical delays and deep divergent histories; simulations do not qualify physical routes or elapsed cryptographic lifetime. Confirm that an offline-recipient output is never reported as independently received before verification.

G5 - Independent security and durability. Obtain independent protocol and implementation review, independent validation implementations and operators, reproducible builds, declared work/security budgets, full-history recovery beyond an issuance era, cryptographic upgrade and long-term archive plans. Resolve release-blocking issues; preserve negative evidence and exact source/profile bindings. A single developer controlling multiple keys is not independent qualification.

G6 - Authenticated release. Publish the executable consensus profile, adoption rules, audited artifacts, supported hardware/resource envelope and operating limits. Only then use a new signed zero-allocation genesis with fresh non-test custody and explicit release authorization. Test balances, test keys and old abandoned networks never migrate. Verify actual distributed packages and services. This publication authorizes no mainnet, real-money activity or automatic upgrade.

G7 - Ordinary-user mining. On predeclared ordinary ARM64 and x86-64 computers, an independent user builds or verifies the package, constructs a local template, chooses a reward key and solo mines with founder, account, hosted API and pool services absent. No prior balance, stake, allowlist or vendor attestation is supplied. Measure valid work, verification cost, CPU/RAM/disk, energy/thermal effects and reward uncertainty under the actual target; include admitted hostile contact load. Test local stop/resume, retained hard-history collapse and early reward concentration. All predeclared devices, attempts, failures and original clocks remain in the denominator. Proof that a binary starts is not evidence of usable participation or profitability.

G8 - Causal payment usefulness. A sender authorizes one exact intent, hands it to willing relays and goes offline. Over a directed path with no return traffic, carry a prefix longer than one batch, the payment and complete dependencies. The destination independently selects and cold-replays its history, observes local eligibility and makes an actual owner-authorized onward payment. Reconnect unequal-work partitions and test surviving common-prefix inputs, removed rewards, conflicting spends, original expiry and chained descendants. Verify exact-signature reconsideration, unchanged reservations and no duplicate credits. Report dependency bytes, waiting/backlog, all pending and failed payments, selected and displaced value, and the user's predeclared acceptance/exposure policy. Merely delivering a capsule, preserving a signature or passing a small easy-work fixture cannot establish useful interstellar settlement.

The complete Survivor-1 candidate contract defines canonical bytes, work and recovery, accounting, branch selection, durable transition behavior, wallet state and delayed transport [6]. G1 maps to T06-T07; G2 to T05/T08-T10; G3 to T01-T04/T11-T13/T17/T22; G4 to T14-T16/T21; G5 to T18-T20 and independent review; G6 to authenticated adoption and release. The additional requirements and D-T01 through D-T08 cases bind G7, G8 and their composition with G1-G6 [7]; existing T01-T22 remain mandatory. The I/S/R/A-G/N/P mapping explicitly records replaced finality/admission obligations and retained security duties. A release may claim only the hardware, work assumptions and route envelopes actually established. No paper, simulation or collection of local passes can substitute for independent qualification or establish irreversible global settlement.

## References


[1] S. Nakamoto. Bitcoin: A Peer-to-Peer Electronic Cash System, 2008. https://bitcoin.org/bitcoin.pdf

[2] NASA. Delay/Disruption Tolerant Networking. https://www.nasa.gov/communicating-with-missions/delay-disruption-tolerant-networking/

[3] S. Burleigh, K. Fall and E. Birrane. Bundle Protocol Version 7, RFC 9171, 2022. https://www.rfc-editor.org/rfc/rfc9171.html

[4] Bitcoin Developer Guide. Block Chain: Proof of Work and Forking. https://developer.bitcoin.org/devguide/block_chain.html

[5] NIST. Module-Lattice-Based Digital Signature Standard, FIPS 204, 2024. https://doi.org/10.6028/NIST.FIPS.204

[6] Rldcoin. Survivor-first protocol specification, RLD-SURVIVOR-1: consensus C01-C11, operations O01-O13, conformance T01-T22 and specification commitments. https://github.com/RunlaiDeng/rldcoin/tree/cf37a9e576f2f6ca581bff0201ce21c9c06cc3ab/docs/protocol/survivor-v1

[7] Rldcoin. Interstellar open-payment requirements: ordinary participation, portable intent, evidence carriage, reconsideration and D-T01 through D-T08. https://github.com/RunlaiDeng/rldcoin/blob/725a5392135862b94446bbb7767699bde7d2f6c4/docs/protocol/INTERSTELLAR_OPEN_PAYMENT_REQUIREMENTS.md

[8] E. Birrane and K. McKeever. Bundle Protocol Security (BPSec), RFC 9172, 2022. https://www.rfc-editor.org/rfc/rfc9172.html

[9] Rldcoin. Interstellar exclusive rights: candidate and security boundary. https://github.com/RunlaiDeng/rldcoin/blob/d9cdf6245609b0bb62f1d1d7f976a212ffd26f4a/docs/research/INTERSTELLAR_EXCLUSIVE_RIGHTS.md